Reversed achievement score in middle school for the oldest child

Current 2022/2023 enrollment defines the middle school sample. One oldest child within this level is selected per household before subject-specific outcome exclusions. 6 minus the recorded response for responses 1 through 5. Higher scores indicate better achievement. Codes 6 and 99 and blanks are excluded separately for each subject. Father estimates are primary and mother estimates are secondary.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.773	0.065 [0.098] N=1,398	0.052 [0.105] N=1,253	0.090 [0.114] N=1,076	0.149 [0.126] N=890	0.156 [0.131] N=713
Social studies and life sciences	3.801	**0.199* [0.105] N=1,392**	**0.191* [0.110] N=1,249**	**0.253** [0.115] N=1,072**	**0.281** [0.126] N=887**	**0.259** [0.128] N=709**
Mathematics	3.586	0.053 [0.160] N=1,398	0.050 [0.167] N=1,253	0.128 [0.182] N=1,076	0.138 [0.197] N=890	0.140 [0.212] N=712
Science	3.764	0.119 [0.112] N=1,388	0.100 [0.117] N=1,245	0.154 [0.125] N=1,069	0.167 [0.138] N=885	0.184 [0.150] N=708
Foreign language	3.554	0.064 [0.122] N=1,385	0.065 [0.131] N=1,243	0.155 [0.134] N=1,069	0.161 [0.146] N=884	0.205 [0.157] N=707
Culture arts and sports	3.943	0.019 [0.110] N=1,299	-0.004 [0.119] N=1,162	0.024 [0.129] N=1,000	0.095 [0.136] N=831	0.047 [0.154] N=669
Vocational and technological	3.829	0.283 [0.173] N=747	0.286 [0.186] N=664	0.274 [0.186] N=573	**0.354* [0.186] N=473**	0.314 [0.206] N=382

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.056 [0.157] N=1,227	-0.089 [0.176] N=1,082	-0.063 [0.199] N=905	0.013 [0.250] N=719	-0.078 [0.266] N=542
Social studies and life sciences	0.164 [0.166] N=1,221	0.154 [0.185] N=1,078	0.274 [0.207] N=901	0.324 [0.242] N=716	0.283 [0.264] N=538
Mathematics	0.028 [0.203] N=1,227	0.042 [0.218] N=1,082	0.205 [0.258] N=905	0.200 [0.302] N=719	0.172 [0.390] N=541
Science	0.074 [0.180] N=1,218	0.050 [0.198] N=1,075	0.176 [0.223] N=899	0.198 [0.267] N=715	0.261 [0.343] N=538
Foreign language	-0.032 [0.184] N=1,214	-0.035 [0.207] N=1,072	0.135 [0.223] N=898	0.090 [0.258] N=713	0.191 [0.315] N=536
Culture arts and sports	0.049 [0.179] N=1,142	0.006 [0.205] N=1,005	0.060 [0.228] N=843	0.176 [0.257] N=674	0.061 [0.331] N=512
Vocational and technological	0.135 [0.172] N=667	0.131 [0.196] N=584	0.115 [0.201] N=493	0.318 [0.224] N=393	0.247 [0.269] N=302

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.773	0.067 [0.096] N=1,398	0.056 [0.103] N=1,253	0.090 [0.112] N=1,076	0.151 [0.124] N=890	0.147 [0.128] N=713
Social studies and life sciences	3.801	**0.199* [0.104] N=1,392**	**0.193* [0.109] N=1,249**	**0.253** [0.114] N=1,072**	**0.282** [0.126] N=887**	**0.253** [0.127] N=709**
Mathematics	3.586	0.052 [0.159] N=1,398	0.048 [0.166] N=1,253	0.128 [0.181] N=1,076	0.137 [0.195] N=890	0.143 [0.211] N=712
Science	3.764	0.119 [0.112] N=1,388	0.101 [0.118] N=1,245	0.154 [0.126] N=1,069	0.168 [0.139] N=885	0.179 [0.150] N=708
Foreign language	3.554	0.069 [0.120] N=1,385	0.073 [0.128] N=1,243	0.155 [0.132] N=1,069	0.164 [0.143] N=884	0.195 [0.153] N=707
Culture arts and sports	3.943	0.019 [0.110] N=1,299	-0.003 [0.119] N=1,162	0.024 [0.129] N=1,000	0.095 [0.135] N=831	0.042 [0.153] N=669
Vocational and technological	3.829	0.284 [0.176] N=747	0.287 [0.189] N=664	0.275 [0.191] N=573	**0.355* [0.190] N=473**	0.320 [0.207] N=382

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.053 [0.153] N=1,227	-0.079 [0.172] N=1,082	-0.057 [0.195] N=905	0.030 [0.247] N=719	-0.062 [0.256] N=542
Social studies and life sciences	0.165 [0.165] N=1,221	0.158 [0.184] N=1,078	0.277 [0.206] N=901	0.332 [0.243] N=716	0.291 [0.260] N=538
Mathematics	0.027 [0.201] N=1,227	0.036 [0.217] N=1,082	0.202 [0.257] N=905	0.188 [0.301] N=719	0.164 [0.389] N=541
Science	0.075 [0.181] N=1,218	0.053 [0.199] N=1,075	0.178 [0.223] N=899	0.206 [0.269] N=715	0.267 [0.343] N=538
Foreign language	-0.023 [0.183] N=1,214	-0.016 [0.206] N=1,072	0.143 [0.222] N=898	0.110 [0.258] N=713	0.199 [0.310] N=536
Culture arts and sports	0.047 [0.180] N=1,142	0.004 [0.205] N=1,005	0.061 [0.228] N=843	0.175 [0.259] N=674	0.064 [0.331] N=512
Vocational and technological	0.139 [0.174] N=667	0.134 [0.197] N=584	0.119 [0.203] N=493	0.322 [0.225] N=393	0.244 [0.269] N=302

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.773	0.066 [0.094] N=1,398	0.055 [0.100] N=1,253	0.086 [0.110] N=1,076	0.148 [0.122] N=890	0.145 [0.126] N=713
Social studies and life sciences	3.801	**0.198* [0.106] N=1,392**	**0.190* [0.111] N=1,249**	**0.244** [0.117] N=1,072**	**0.272** [0.128] N=887**	**0.247* [0.130] N=709**
Mathematics	3.586	0.052 [0.166] N=1,398	0.045 [0.172] N=1,253	0.117 [0.188] N=1,076	0.120 [0.203] N=890	0.130 [0.221] N=712
Science	3.764	0.121 [0.115] N=1,388	0.101 [0.120] N=1,245	0.147 [0.128] N=1,069	0.161 [0.140] N=885	0.169 [0.153] N=708
Foreign language	3.554	0.059 [0.122] N=1,385	0.059 [0.130] N=1,243	0.137 [0.136] N=1,069	0.145 [0.147] N=884	0.184 [0.156] N=707
Culture arts and sports	3.943	0.024 [0.112] N=1,299	0.001 [0.121] N=1,162	0.021 [0.129] N=1,000	0.092 [0.136] N=831	0.038 [0.153] N=669
Vocational and technological	3.829	0.267 [0.184] N=747	0.269 [0.199] N=664	0.250 [0.200] N=573	**0.343* [0.198] N=473**	0.330 [0.217] N=382

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.051 [0.144] N=1,227	-0.079 [0.163] N=1,082	-0.072 [0.185] N=905	0.013 [0.236] N=719	-0.109 [0.247] N=542
Social studies and life sciences	0.156 [0.160] N=1,221	0.142 [0.177] N=1,078	0.246 [0.200] N=901	0.294 [0.233] N=716	0.248 [0.246] N=538
Mathematics	0.006 [0.193] N=1,227	0.003 [0.204] N=1,082	0.138 [0.241] N=905	0.112 [0.282] N=719	0.088 [0.365] N=541
Science	0.081 [0.173] N=1,218	0.056 [0.188] N=1,075	0.169 [0.212] N=899	0.205 [0.258] N=715	0.257 [0.329] N=538
Foreign language	-0.037 [0.179] N=1,214	-0.042 [0.199] N=1,072	0.092 [0.216] N=898	0.032 [0.253] N=713	0.138 [0.301] N=536
Culture arts and sports	0.057 [0.170] N=1,142	0.013 [0.195] N=1,005	0.064 [0.210] N=843	0.178 [0.241] N=674	0.066 [0.306] N=512
Vocational and technological	0.126 [0.177] N=667	0.101 [0.198] N=584	0.077 [0.209] N=493	0.344 [0.235] N=393	0.283 [0.278] N=302

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.773	0.066 [0.096] N=1,398	0.058 [0.103] N=1,253	0.089 [0.113] N=1,076	0.140 [0.124] N=890	0.134 [0.128] N=713
Social studies and life sciences	3.801	**0.201* [0.106] N=1,392**	**0.201* [0.112] N=1,249**	**0.255** [0.118] N=1,072**	**0.271** [0.128] N=887**	**0.239* [0.129] N=709**
Mathematics	3.586	0.053 [0.163] N=1,398	0.056 [0.169] N=1,253	0.126 [0.183] N=1,076	0.120 [0.198] N=890	0.122 [0.215] N=712
Science	3.764	0.118 [0.113] N=1,388	0.105 [0.119] N=1,245	0.148 [0.128] N=1,069	0.152 [0.140] N=885	0.160 [0.151] N=708
Foreign language	3.554	0.067 [0.124] N=1,385	0.076 [0.132] N=1,243	0.149 [0.136] N=1,069	0.151 [0.147] N=884	0.181 [0.157] N=707
Culture arts and sports	3.943	0.013 [0.110] N=1,299	-0.005 [0.119] N=1,162	0.009 [0.128] N=1,000	0.072 [0.134] N=831	0.004 [0.150] N=669
Vocational and technological	3.829	0.282 [0.181] N=747	0.279 [0.194] N=664	0.261 [0.195] N=573	**0.341* [0.193] N=473**	0.299 [0.206] N=382

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.062 [0.148] N=1,227	-0.081 [0.168] N=1,082	-0.069 [0.191] N=905	0.009 [0.242] N=719	-0.090 [0.247] N=542
Social studies and life sciences	0.150 [0.162] N=1,221	0.152 [0.184] N=1,078	0.255 [0.205] N=901	0.291 [0.237] N=716	0.268 [0.240] N=538
Mathematics	0.006 [0.194] N=1,227	0.029 [0.208] N=1,082	0.174 [0.243] N=905	0.123 [0.281] N=719	0.112 [0.349] N=541
Science	0.061 [0.170] N=1,218	0.051 [0.189] N=1,075	0.156 [0.213] N=899	0.171 [0.256] N=715	0.249 [0.316] N=538
Foreign language	-0.049 [0.182] N=1,214	-0.031 [0.204] N=1,072	0.105 [0.220] N=898	0.051 [0.255] N=713	0.153 [0.293] N=536
Culture arts and sports	0.042 [0.169] N=1,142	0.010 [0.193] N=1,005	0.055 [0.211] N=843	0.170 [0.236] N=674	0.064 [0.290] N=512
Vocational and technological	0.122 [0.175] N=667	0.100 [0.196] N=584	0.076 [0.206] N=493	0.307 [0.234] N=393	0.241 [0.269] N=302

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.762	0.083 [0.097] N=1,327	0.077 [0.104] N=1,189	0.122 [0.110] N=1,021	0.187 [0.120] N=847	0.136 [0.124] N=678
Social studies and life sciences	3.789	**0.219** [0.100] N=1,321**	**0.216** [0.105] N=1,185**	**0.292*** [0.106] N=1,017**	**0.325*** [0.114] N=844**	**0.248** [0.111] N=674**
Mathematics	3.571	0.083 [0.151] N=1,327	0.081 [0.158] N=1,189	0.176 [0.168] N=1,021	0.206 [0.183] N=847	0.208 [0.192] N=677
Science	3.751	0.125 [0.111] N=1,317	0.115 [0.116] N=1,181	0.189 [0.120] N=1,014	0.208 [0.133] N=842	0.187 [0.145] N=673
Foreign language	3.539	0.077 [0.117] N=1,315	0.089 [0.125] N=1,180	**0.208* [0.125] N=1,015**	0.222 [0.134] N=842	0.203 [0.146] N=673
Culture arts and sports	3.935	0.065 [0.100] N=1,247	0.050 [0.110] N=1,114	0.081 [0.121] N=960	0.153 [0.123] N=797	0.108 [0.140] N=642
Vocational and technological	3.825	**0.289* [0.172] N=712**	0.298 [0.188] N=633	0.295 [0.192] N=547	**0.374* [0.193] N=451**	0.342 [0.209] N=363

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.016 [0.164] N=1,167	-0.034 [0.183] N=1,029	-0.002 [0.209] N=861	0.091 [0.263] N=687	-0.113 [0.266] N=518
Social studies and life sciences	0.194 [0.167] N=1,161	0.186 [0.186] N=1,025	0.323 [0.209] N=857	0.390 [0.247] N=684	0.258 [0.256] N=514
Mathematics	0.037 [0.196] N=1,167	0.044 [0.213] N=1,029	0.220 [0.250] N=861	0.215 [0.303] N=687	0.174 [0.379] N=517
Science	0.099 [0.182] N=1,158	0.087 [0.200] N=1,022	0.242 [0.224] N=855	0.289 [0.278] N=683	0.311 [0.349] N=514
Foreign language	0.001 [0.178] N=1,155	0.019 [0.202] N=1,020	0.235 [0.213] N=855	0.225 [0.248] N=682	0.246 [0.303] N=513
Culture arts and sports	0.074 [0.174] N=1,098	0.035 [0.199] N=965	0.092 [0.226] N=811	0.202 [0.255] N=648	0.089 [0.336] N=493
Vocational and technological	0.123 [0.171] N=636	0.126 [0.193] N=557	0.131 [0.194] N=471	0.314 [0.223] N=375	0.253 [0.269] N=287

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.762	0.085 [0.097] N=1,327	0.075 [0.103] N=1,189	0.114 [0.110] N=1,021	0.179 [0.120] N=847	0.125 [0.124] N=678
Social studies and life sciences	3.789	**0.220** [0.100] N=1,321**	**0.213** [0.105] N=1,185**	**0.284*** [0.106] N=1,017**	**0.318*** [0.115] N=844**	**0.242** [0.113] N=674**
Mathematics	3.571	0.083 [0.152] N=1,327	0.077 [0.159] N=1,189	0.167 [0.171] N=1,021	0.196 [0.188] N=847	0.190 [0.198] N=677
Science	3.751	0.126 [0.111] N=1,317	0.112 [0.117] N=1,181	0.179 [0.121] N=1,014	0.201 [0.134] N=842	0.181 [0.147] N=673
Foreign language	3.539	0.075 [0.118] N=1,315	0.081 [0.126] N=1,180	0.192 [0.126] N=1,015	0.205 [0.136] N=842	0.184 [0.149] N=673
Culture arts and sports	3.935	0.065 [0.102] N=1,247	0.046 [0.111] N=1,114	0.071 [0.122] N=960	0.144 [0.125] N=797	0.093 [0.142] N=642
Vocational and technological	3.825	**0.307* [0.177] N=712**	0.310 [0.193] N=633	0.305 [0.196] N=547	**0.388* [0.199] N=451**	0.348 [0.215] N=363

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.012 [0.166] N=1,167	-0.029 [0.185] N=1,029	0.003 [0.211] N=861	0.100 [0.265] N=687	-0.109 [0.266] N=518
Social studies and life sciences	0.196 [0.167] N=1,161	0.191 [0.189] N=1,025	0.326 [0.211] N=857	0.396 [0.247] N=684	0.264 [0.259] N=514
Mathematics	0.044 [0.198] N=1,167	0.050 [0.215] N=1,029	0.225 [0.254] N=861	0.226 [0.308] N=687	0.181 [0.387] N=517
Science	0.103 [0.182] N=1,158	0.092 [0.202] N=1,022	0.245 [0.227] N=855	0.299 [0.280] N=683	0.320 [0.351] N=514
Foreign language	0.009 [0.178] N=1,155	0.026 [0.203] N=1,020	0.239 [0.214] N=855	0.236 [0.250] N=682	0.250 [0.307] N=513
Culture arts and sports	0.084 [0.178] N=1,098	0.046 [0.204] N=965	0.100 [0.229] N=811	0.219 [0.257] N=648	0.094 [0.335] N=493
Vocational and technological	0.137 [0.172] N=636	0.132 [0.194] N=557	0.135 [0.195] N=471	0.334 [0.225] N=375	0.249 [0.268] N=287

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.790	0.043 [0.072] N=2,021	0.012 [0.074] N=1,876	0.034 [0.081] N=1,668	0.055 [0.087] N=1,456	0.037 [0.093] N=1,195
Social studies and life sciences	3.818	-0.016 [0.068] N=2,013	-0.043 [0.073] N=1,868	0.011 [0.078] N=1,662	0.014 [0.084] N=1,452	0.011 [0.093] N=1,191
Mathematics	3.598	-0.053 [0.074] N=2,022	-0.107 [0.078] N=1,878	-0.046 [0.081] N=1,670	-0.009 [0.090] N=1,458	0.038 [0.098] N=1,197
Science	3.798	0.014 [0.072] N=2,008	-0.024 [0.074] N=1,865	0.036 [0.078] N=1,659	0.057 [0.083] N=1,450	0.121 [0.087] N=1,190
Foreign language	3.551	0.072 [0.085] N=2,001	0.039 [0.090] N=1,857	0.081 [0.096] N=1,650	0.126 [0.107] N=1,440	0.155 [0.116] N=1,182
Culture arts and sports	3.966	0.027 [0.070] N=1,878	-0.000 [0.073] N=1,744	0.005 [0.079] N=1,551	-0.006 [0.088] N=1,353	0.058 [0.099] N=1,118
Vocational and technological	3.838	-0.023 [0.085] N=1,098	-0.014 [0.089] N=1,013	0.002 [0.093] N=909	0.001 [0.104] N=796	0.115 [0.110] N=660

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.121 [0.091] N=1,693	0.060 [0.098] N=1,548	0.110 [0.110] N=1,340	0.172 [0.134] N=1,128	0.171 [0.165] N=867
Social studies and life sciences	0.013 [0.080] N=1,688	-0.040 [0.087] N=1,543	0.060 [0.097] N=1,337	0.063 [0.112] N=1,127	0.056 [0.131] N=866
Mathematics	-0.055 [0.106] N=1,694	-0.162 [0.117] N=1,550	-0.061 [0.127] N=1,342	-0.002 [0.147] N=1,130	0.096 [0.172] N=869
Science	0.015 [0.102] N=1,684	-0.071 [0.107] N=1,541	0.027 [0.118] N=1,335	0.044 [0.138] N=1,126	0.160 [0.163] N=866
Foreign language	0.042 [0.115] N=1,678	-0.042 [0.127] N=1,534	0.010 [0.145] N=1,327	0.089 [0.179] N=1,117	0.183 [0.225] N=859
Culture arts and sports	0.010 [0.083] N=1,574	-0.040 [0.085] N=1,440	-0.046 [0.092] N=1,247	-0.083 [0.118] N=1,049	0.022 [0.143] N=814
Vocational and technological	-0.033 [0.115] N=929	-0.013 [0.119] N=844	-0.008 [0.129] N=740	-0.028 [0.161] N=627	0.200 [0.191] N=491

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.790	0.036 [0.072] N=2,021	-0.000 [0.074] N=1,876	0.021 [0.080] N=1,668	0.038 [0.087] N=1,456	0.024 [0.093] N=1,195
Social studies and life sciences	3.818	-0.019 [0.068] N=2,013	-0.047 [0.072] N=1,868	0.006 [0.077] N=1,662	0.006 [0.083] N=1,452	0.004 [0.092] N=1,191
Mathematics	3.598	-0.050 [0.074] N=2,022	-0.102 [0.078] N=1,878	-0.039 [0.081] N=1,670	-0.002 [0.089] N=1,458	0.046 [0.097] N=1,197
Science	3.798	0.013 [0.072] N=2,008	-0.027 [0.074] N=1,865	0.034 [0.078] N=1,659	0.053 [0.083] N=1,450	0.119 [0.087] N=1,190
Foreign language	3.551	0.064 [0.086] N=2,001	0.027 [0.090] N=1,857	0.070 [0.096] N=1,650	0.111 [0.107] N=1,440	0.146 [0.117] N=1,182
Culture arts and sports	3.966	0.025 [0.071] N=1,878	-0.003 [0.073] N=1,744	0.002 [0.079] N=1,551	-0.009 [0.088] N=1,353	0.054 [0.099] N=1,118
Vocational and technological	3.838	-0.022 [0.085] N=1,098	-0.011 [0.089] N=1,013	0.006 [0.091] N=909	0.008 [0.103] N=796	0.119 [0.110] N=660

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.116 [0.092] N=1,693	0.047 [0.098] N=1,548	0.094 [0.110] N=1,340	0.144 [0.133] N=1,128	0.150 [0.166] N=867
Social studies and life sciences	0.011 [0.080] N=1,688	-0.044 [0.088] N=1,543	0.056 [0.097] N=1,337	0.054 [0.112] N=1,127	0.049 [0.131] N=866
Mathematics	-0.052 [0.106] N=1,694	-0.155 [0.118] N=1,550	-0.053 [0.128] N=1,342	0.009 [0.148] N=1,130	0.111 [0.172] N=869
Science	0.014 [0.102] N=1,684	-0.073 [0.108] N=1,541	0.025 [0.118] N=1,335	0.038 [0.138] N=1,126	0.160 [0.163] N=866
Foreign language	0.035 [0.114] N=1,678	-0.059 [0.126] N=1,534	-0.007 [0.143] N=1,327	0.060 [0.177] N=1,117	0.163 [0.226] N=859
Culture arts and sports	0.009 [0.082] N=1,574	-0.044 [0.084] N=1,440	-0.049 [0.092] N=1,247	-0.090 [0.118] N=1,049	0.013 [0.143] N=814
Vocational and technological	-0.033 [0.115] N=929	-0.012 [0.119] N=844	-0.011 [0.128] N=740	-0.023 [0.160] N=627	0.201 [0.192] N=491

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.790	0.031 [0.072] N=2,021	-0.005 [0.074] N=1,876	0.015 [0.080] N=1,668	0.038 [0.087] N=1,456	0.017 [0.093] N=1,195
Social studies and life sciences	3.818	-0.026 [0.067] N=2,013	-0.054 [0.071] N=1,868	-0.002 [0.077] N=1,662	0.004 [0.083] N=1,452	-0.004 [0.093] N=1,191
Mathematics	3.598	-0.056 [0.074] N=2,022	-0.110 [0.078] N=1,878	-0.050 [0.081] N=1,670	-0.003 [0.088] N=1,458	0.037 [0.098] N=1,197
Science	3.798	0.004 [0.071] N=2,008	-0.035 [0.073] N=1,865	0.023 [0.078] N=1,659	0.049 [0.083] N=1,450	0.108 [0.088] N=1,190
Foreign language	3.551	0.054 [0.084] N=2,001	0.014 [0.089] N=1,857	0.051 [0.095] N=1,650	0.103 [0.105] N=1,440	0.126 [0.115] N=1,182
Culture arts and sports	3.966	0.026 [0.069] N=1,878	-0.004 [0.072] N=1,744	0.002 [0.078] N=1,551	-0.005 [0.087] N=1,353	0.051 [0.098] N=1,118
Vocational and technological	3.838	-0.031 [0.083] N=1,098	-0.022 [0.087] N=1,013	-0.009 [0.091] N=909	-0.009 [0.103] N=796	0.085 [0.112] N=660

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.116 [0.092] N=1,693	0.047 [0.098] N=1,548	0.091 [0.111] N=1,340	0.153 [0.134] N=1,128	0.157 [0.169] N=867
Social studies and life sciences	0.013 [0.078] N=1,688	-0.043 [0.085] N=1,543	0.053 [0.096] N=1,337	0.062 [0.112] N=1,127	0.054 [0.132] N=866
Mathematics	-0.047 [0.106] N=1,694	-0.155 [0.117] N=1,550	-0.058 [0.127] N=1,342	0.028 [0.145] N=1,130	0.124 [0.168] N=869
Science	0.017 [0.101] N=1,684	-0.071 [0.106] N=1,541	0.022 [0.118] N=1,335	0.057 [0.137] N=1,126	0.169 [0.163] N=866
Foreign language	0.041 [0.111] N=1,678	-0.057 [0.122] N=1,534	-0.013 [0.137] N=1,327	0.085 [0.169] N=1,117	0.183 [0.210] N=859
Culture arts and sports	0.018 [0.079] N=1,574	-0.039 [0.081] N=1,440	-0.045 [0.088] N=1,247	-0.067 [0.113] N=1,049	0.033 [0.135] N=814
Vocational and technological	-0.011 [0.114] N=929	0.008 [0.117] N=844	0.001 [0.127] N=740	-0.014 [0.157] N=627	0.210 [0.195] N=491

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.790	0.036 [0.073] N=2,021	-0.001 [0.075] N=1,876	0.018 [0.081] N=1,668	0.043 [0.088] N=1,456	0.024 [0.095] N=1,195
Social studies and life sciences	3.818	-0.022 [0.068] N=2,013	-0.051 [0.072] N=1,868	-0.001 [0.079] N=1,662	0.009 [0.085] N=1,452	0.003 [0.094] N=1,191
Mathematics	3.598	-0.052 [0.075] N=2,022	-0.105 [0.079] N=1,878	-0.047 [0.083] N=1,670	0.002 [0.090] N=1,458	0.044 [0.099] N=1,197
Science	3.798	0.010 [0.073] N=2,008	-0.031 [0.075] N=1,865	0.028 [0.080] N=1,659	0.059 [0.086] N=1,450	0.116 [0.090] N=1,190
Foreign language	3.551	0.061 [0.085] N=2,001	0.020 [0.089] N=1,857	0.054 [0.096] N=1,650	0.109 [0.107] N=1,440	0.137 [0.117] N=1,182
Culture arts and sports	3.966	0.026 [0.068] N=1,878	-0.004 [0.070] N=1,744	0.000 [0.076] N=1,551	-0.001 [0.087] N=1,353	0.051 [0.097] N=1,118
Vocational and technological	3.838	-0.022 [0.086] N=1,098	-0.014 [0.090] N=1,013	-0.005 [0.094] N=909	0.003 [0.106] N=796	0.099 [0.115] N=660

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.121 [0.094] N=1,693	0.049 [0.100] N=1,548	0.090 [0.112] N=1,340	0.155 [0.137] N=1,128	0.155 [0.173] N=867
Social studies and life sciences	0.014 [0.080] N=1,688	-0.046 [0.087] N=1,543	0.048 [0.098] N=1,337	0.066 [0.115] N=1,127	0.061 [0.135] N=866
Mathematics	-0.052 [0.106] N=1,694	-0.157 [0.117] N=1,550	-0.064 [0.129] N=1,342	0.021 [0.146] N=1,130	0.122 [0.170] N=869
Science	0.020 [0.104] N=1,684	-0.072 [0.109] N=1,541	0.024 [0.121] N=1,335	0.066 [0.143] N=1,126	0.176 [0.170] N=866
Foreign language	0.043 [0.111] N=1,678	-0.059 [0.121] N=1,534	-0.023 [0.136] N=1,327	0.076 [0.171] N=1,117	0.185 [0.215] N=859
Culture arts and sports	0.018 [0.080] N=1,574	-0.038 [0.080] N=1,440	-0.044 [0.087] N=1,247	-0.056 [0.114] N=1,049	0.035 [0.136] N=814
Vocational and technological	-0.003 [0.116] N=929	0.016 [0.119] N=844	0.009 [0.130] N=740	0.012 [0.165] N=627	0.247 [0.193] N=491

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.779	0.057 [0.070] N=1,910	0.029 [0.072] N=1,775	0.057 [0.076] N=1,573	0.072 [0.082] N=1,372	0.057 [0.087] N=1,131
Social studies and life sciences	3.803	0.011 [0.068] N=1,901	-0.009 [0.072] N=1,766	0.048 [0.076] N=1,566	0.040 [0.081] N=1,367	0.036 [0.090] N=1,126
Mathematics	3.588	-0.007 [0.075] N=1,910	-0.056 [0.079] N=1,776	0.010 [0.085] N=1,574	0.055 [0.094] N=1,373	0.077 [0.102] N=1,132
Science	3.786	0.044 [0.074] N=1,896	0.010 [0.075] N=1,763	0.073 [0.080] N=1,563	0.086 [0.084] N=1,365	0.132 [0.089] N=1,125
Foreign language	3.540	0.104 [0.086] N=1,892	0.069 [0.091] N=1,758	0.121 [0.097] N=1,557	0.149 [0.110] N=1,358	0.165 [0.119] N=1,119
Culture arts and sports	3.959	0.040 [0.068] N=1,791	0.012 [0.070] N=1,665	0.026 [0.075] N=1,477	0.012 [0.084] N=1,288	0.055 [0.093] N=1,068
Vocational and technological	3.839	-0.017 [0.086] N=1,042	-0.019 [0.091] N=962	-0.003 [0.095] N=860	-0.016 [0.106] N=752	0.066 [0.114] N=626

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.118 [0.091] N=1,598	0.061 [0.096] N=1,463	0.117 [0.108] N=1,261	0.151 [0.131] N=1,060	0.119 [0.162] N=819
Social studies and life sciences	0.033 [0.079] N=1,592	-0.007 [0.086] N=1,457	0.097 [0.095] N=1,257	0.073 [0.112] N=1,058	0.042 [0.131] N=817
Mathematics	0.004 [0.101] N=1,598	-0.089 [0.113] N=1,464	0.031 [0.122] N=1,262	0.109 [0.142] N=1,061	0.143 [0.162] N=820
Science	0.058 [0.103] N=1,588	-0.016 [0.108] N=1,455	0.094 [0.118] N=1,255	0.094 [0.138] N=1,057	0.147 [0.164] N=817
Foreign language	0.076 [0.115] N=1,585	-0.007 [0.127] N=1,451	0.060 [0.143] N=1,250	0.104 [0.182] N=1,051	0.133 [0.222] N=812
Culture arts and sports	0.018 [0.081] N=1,498	-0.036 [0.082] N=1,372	-0.030 [0.091] N=1,184	-0.078 [0.119] N=995	-0.033 [0.138] N=775
Vocational and technological	-0.018 [0.118] N=878	-0.020 [0.124] N=798	-0.011 [0.134] N=696	-0.055 [0.167] N=588	0.086 [0.211] N=462

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.779	0.064 [0.070] N=1,910	0.034 [0.072] N=1,775	0.061 [0.076] N=1,573	0.079 [0.082] N=1,372	0.063 [0.087] N=1,131
Social studies and life sciences	3.803	0.013 [0.068] N=1,901	-0.008 [0.072] N=1,766	0.047 [0.077] N=1,566	0.045 [0.082] N=1,367	0.042 [0.091] N=1,126
Mathematics	3.588	0.000 [0.075] N=1,910	-0.051 [0.079] N=1,776	0.014 [0.084] N=1,574	0.062 [0.093] N=1,373	0.085 [0.101] N=1,132
Science	3.786	0.050 [0.073] N=1,896	0.014 [0.075] N=1,763	0.077 [0.079] N=1,563	0.097 [0.084] N=1,365	0.144 [0.088] N=1,125
Foreign language	3.540	0.110 [0.085] N=1,892	0.073 [0.090] N=1,758	0.123 [0.095] N=1,557	0.159 [0.108] N=1,358	0.174 [0.116] N=1,119
Culture arts and sports	3.959	0.039 [0.068] N=1,791	0.009 [0.071] N=1,665	0.019 [0.076] N=1,477	0.009 [0.085] N=1,288	0.056 [0.094] N=1,068
Vocational and technological	3.839	-0.010 [0.086] N=1,042	-0.012 [0.091] N=962	0.009 [0.093] N=860	-0.002 [0.104] N=752	0.089 [0.110] N=626

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.123 [0.090] N=1,598	0.060 [0.096] N=1,463	0.108 [0.107] N=1,261	0.144 [0.131] N=1,060	0.114 [0.162] N=819
Social studies and life sciences	0.031 [0.080] N=1,592	-0.012 [0.087] N=1,457	0.086 [0.096] N=1,257	0.065 [0.114] N=1,058	0.041 [0.132] N=817
Mathematics	0.007 [0.101] N=1,598	-0.092 [0.113] N=1,464	0.017 [0.124] N=1,262	0.103 [0.144] N=1,061	0.143 [0.165] N=820
Science	0.057 [0.102] N=1,588	-0.023 [0.108] N=1,455	0.080 [0.117] N=1,255	0.085 [0.139] N=1,057	0.143 [0.163] N=817
Foreign language	0.077 [0.115] N=1,585	-0.011 [0.127] N=1,451	0.049 [0.143] N=1,250	0.101 [0.182] N=1,051	0.132 [0.223] N=812
Culture arts and sports	0.015 [0.082] N=1,498	-0.042 [0.084] N=1,372	-0.044 [0.093] N=1,184	-0.090 [0.121] N=995	-0.037 [0.141] N=775
Vocational and technological	-0.019 [0.119] N=878	-0.024 [0.124] N=798	-0.012 [0.134] N=696	-0.061 [0.167] N=588	0.082 [0.208] N=462

Notes: Cells report the reform eligibility coefficient, parental birth-month-clustered standard error in brackets, and N. All models use exact child-month fixed effects calculated on 16 December 2022, split linear trends at January 1987, and FAKTOR_FERT child survey weights. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add categorical aid count and any aid. Aid specifications require valid yes/no answers to all six household aid questions. Income and aid controls are separate alternatives. Columns use 8- to 4-year bandwidths. Donuts omit running months -12 through +11. No child-age interval restriction is imposed; complete birth dates are needed for sibling selection and monthly controls. P is the unweighted positive-case count in the no-donut 8-year sample. NA denotes a failed or zero-standard-error fit. These are caregiver assessments. School-level and subject-response selection can affect interpretation. *** p<0.01, ** p<0.05, * p<0.10.